# A Ring of Lights

Eric Shen

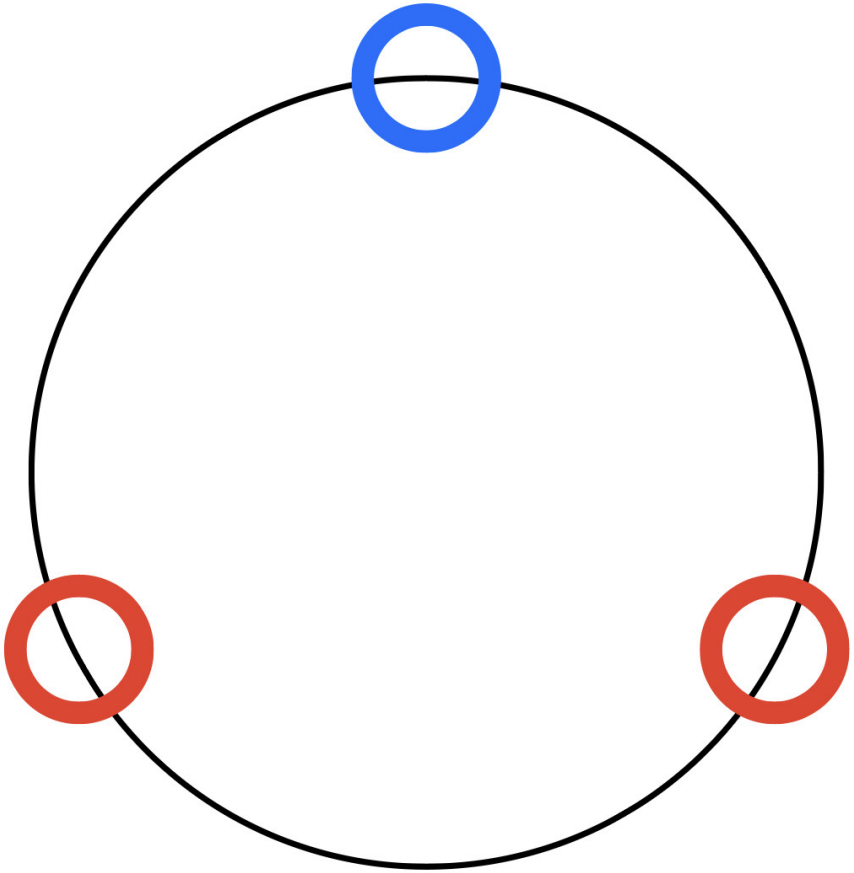
October 2022

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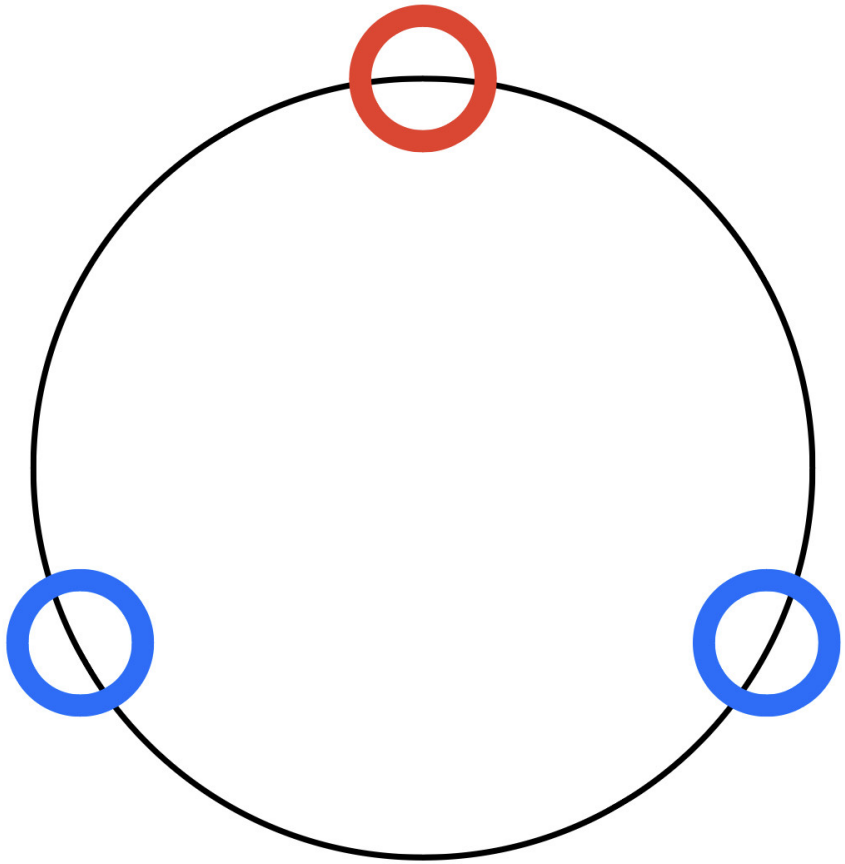
**Problem**

*There are $n > 1$ lamps arranged (evenly spaced) in a circle. Initially, one of them is turned on, and the rest are off. It is permitted to choose any regular polygon whose vertices are lamps and toggle all of their states simultaneously. For which positive integers $n$ is it possible to turn all the lamps off after a finite number of such operations?*

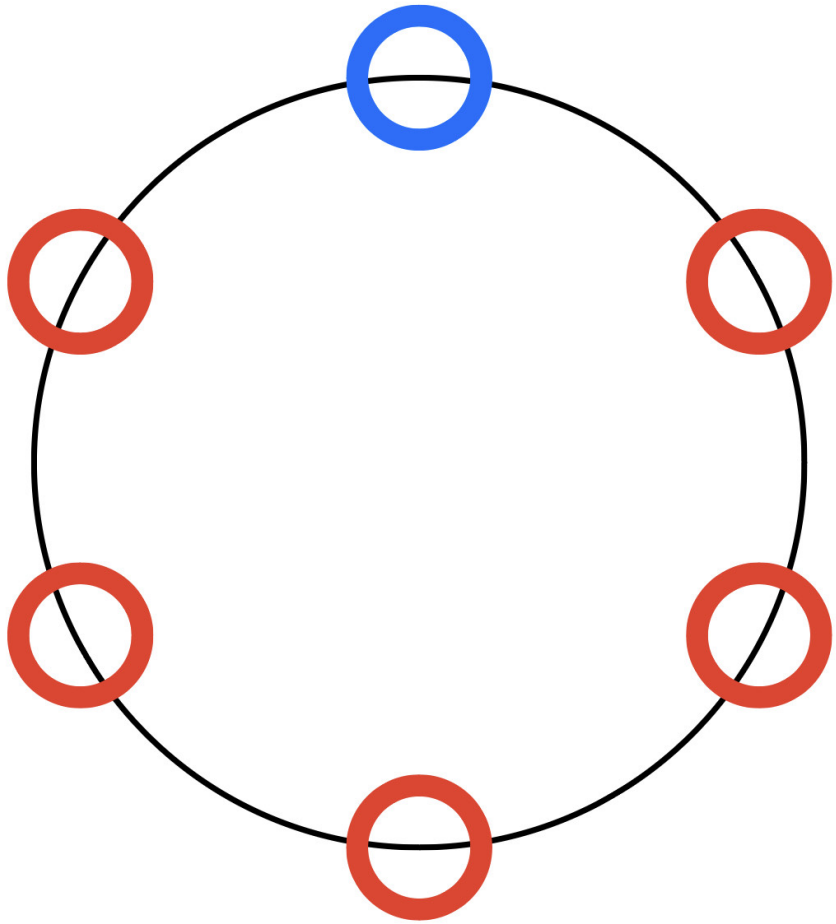
# Preliminary Remarks



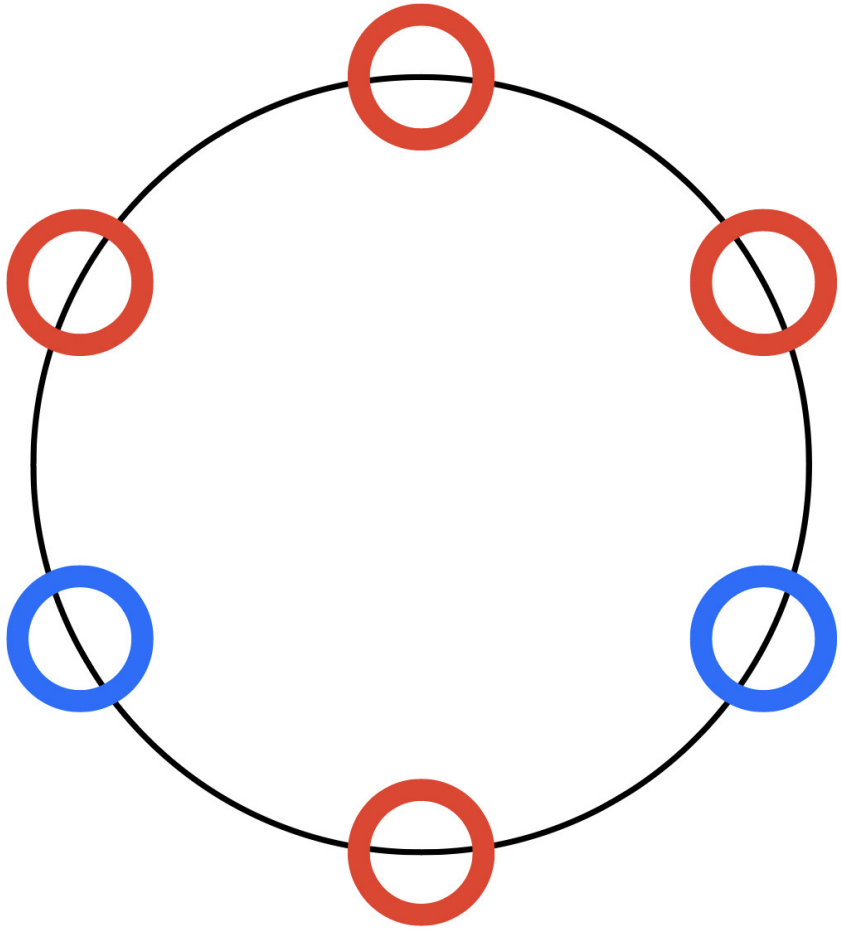
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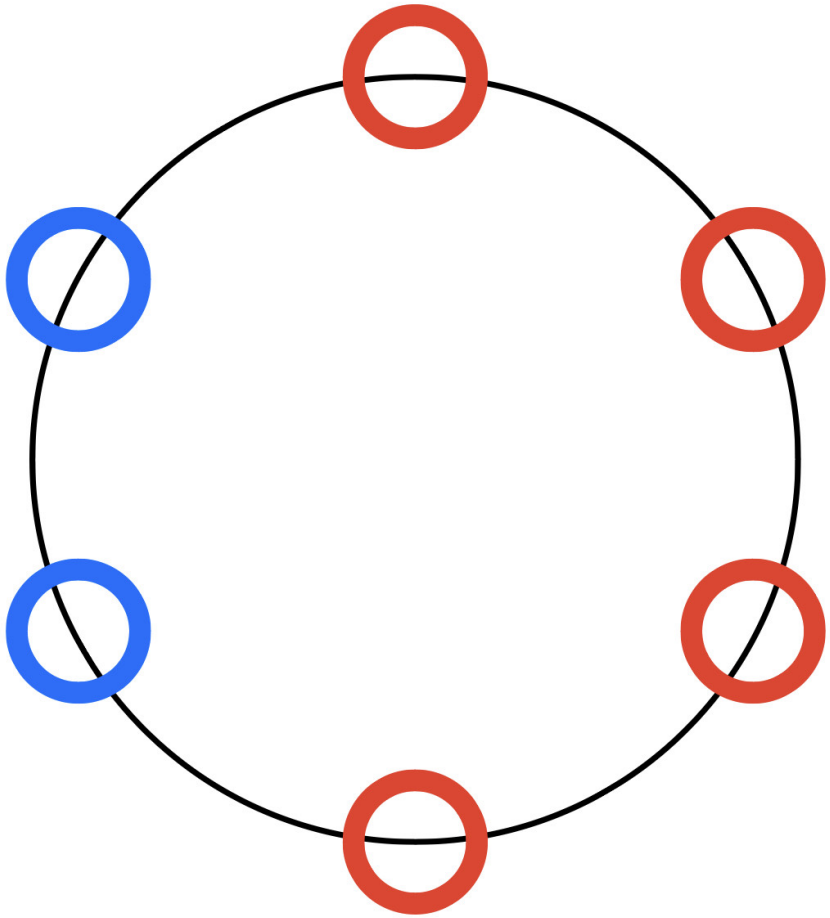
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*Two opposite corners of an* $8 \times 8$ *chessboard are taken out. Can we still cover the chessboard with dominoes?*

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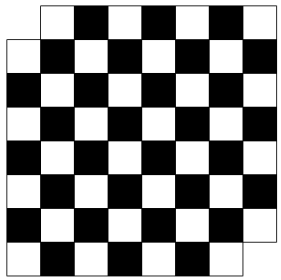
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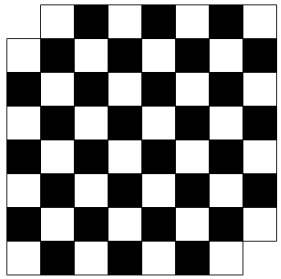
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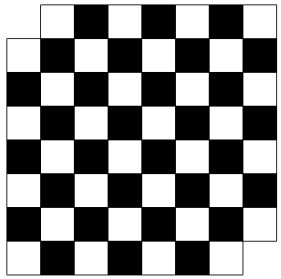
## Proof.

Color the chessboard with a checkerboard pattern. Each domino always lies over one black and one white square. But we don't have the same number of black and white squares, so we can't cover the chessboard. □

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Lesson: a good invariant can make analyzing a "big" problem much easier.

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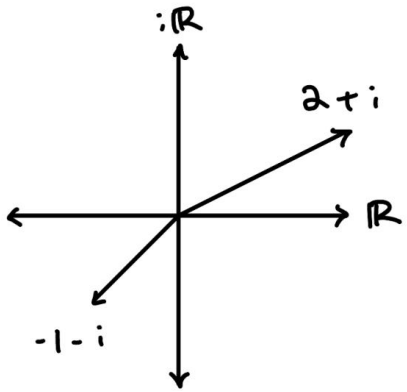
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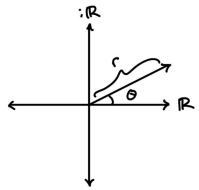
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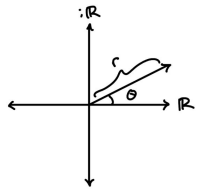
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This graph interpretation is especially helpful because we can write complex numbers in terms of their distance from the origin (the magnitude), along with the angle they make from the real axis (the argument).



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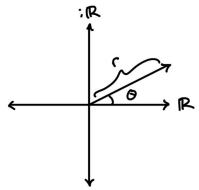
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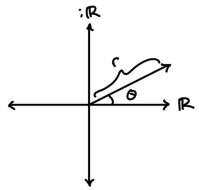

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**Theorem (De Moivre's)**

*Given any two complex numbers $z_1, z_2$, the magnitude of their product $z_1 z_2$ is the product of the magnitudes of $z_1$ and $z_2$, and the argument of $z_1 z_2$ is the sum of the arguments of $z_1$ and $z_2$.*

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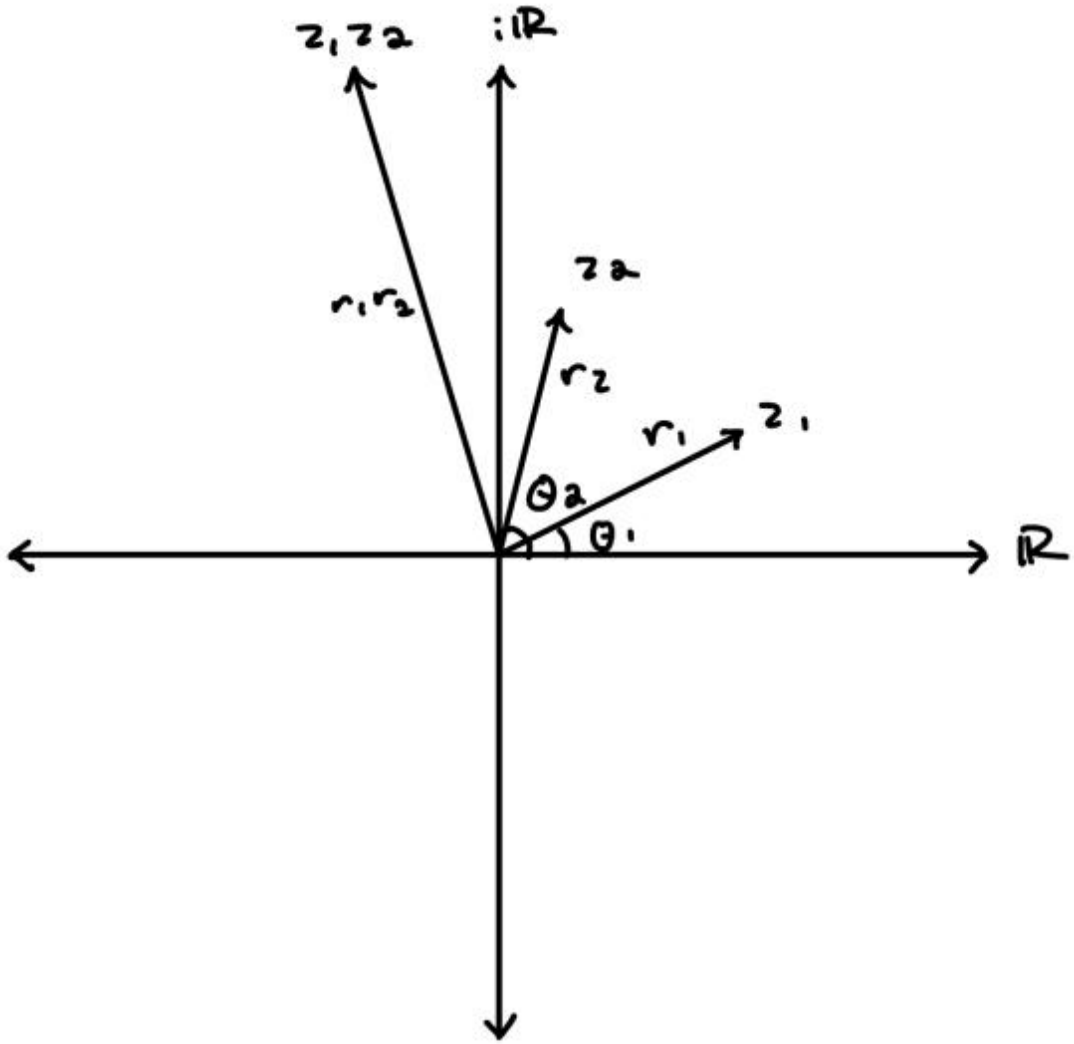

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This makes it geometrically very intuitive to multiply complex numbers. Things are especially easy when all of the magnitudes we are looking at are 1, and so multiplying complex numbers in that case just amounts to adding angles.

# Multiplying Complex Numbers



# Roots of Unity

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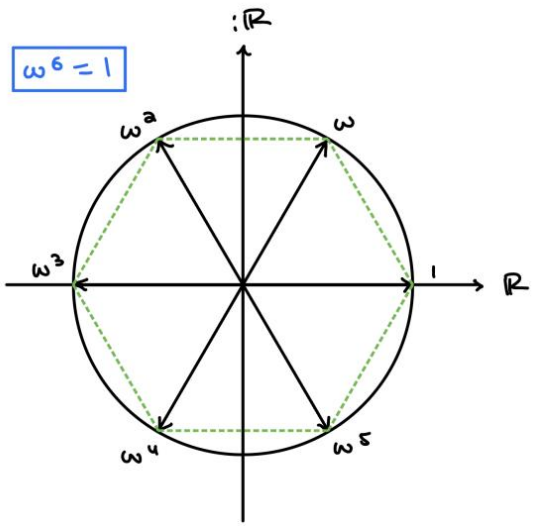
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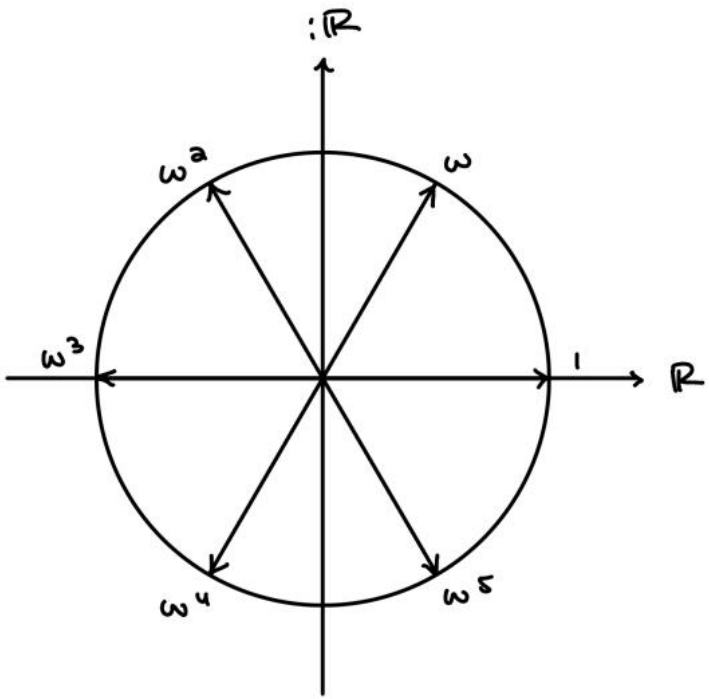
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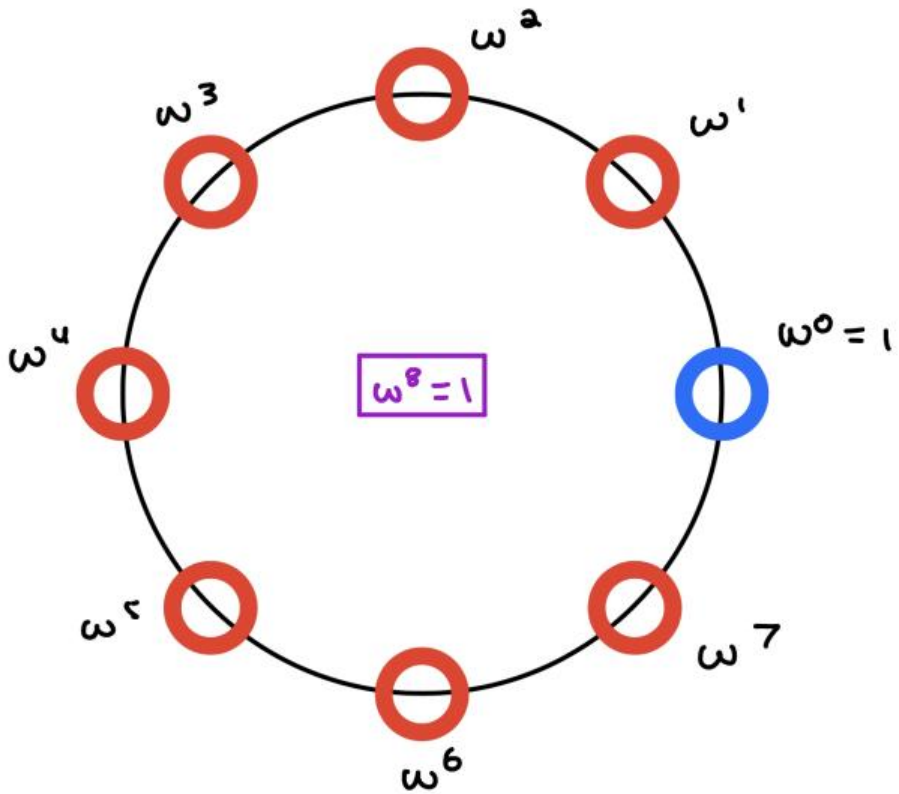


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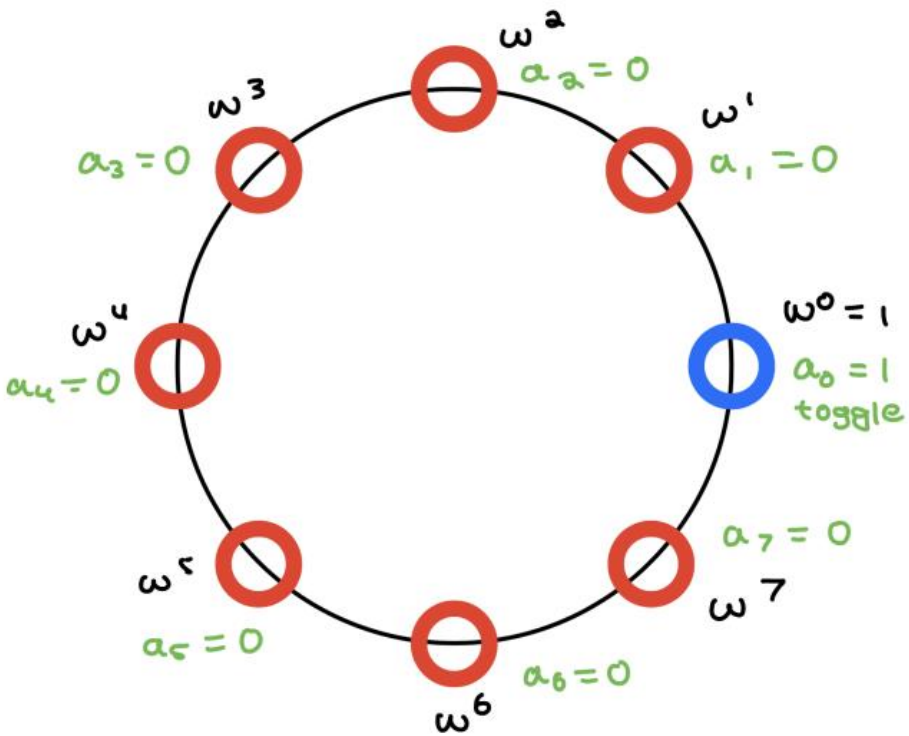


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We'll also label the vertices with the number of times they have been toggled (with the vertex corr. to $\omega^0$ starting off with having been toggled once.)

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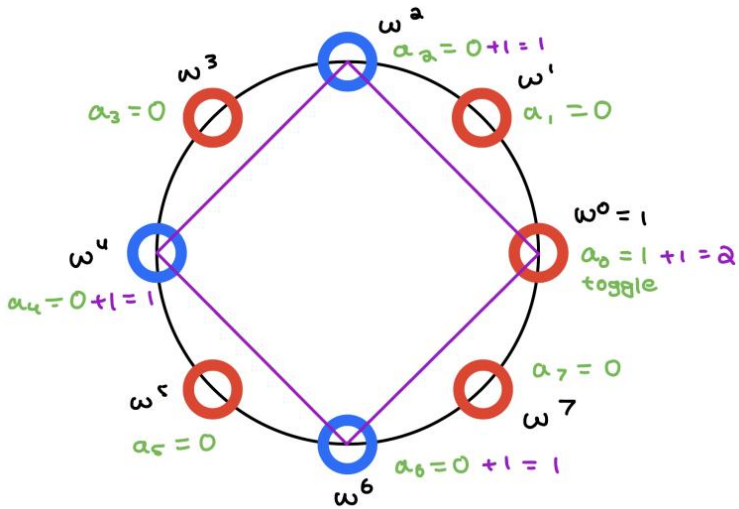


# How do we use this?

Evidently we have that the $n$th vertex is toggled *on* iff $a_n$ is odd. Moreover, we have that each time we toggle a regular polygon, we add one to $a_x, a_{x+d}, \ldots a_{x+n-d}$ for some integer $x$ and integer $d$ dividing $n$.

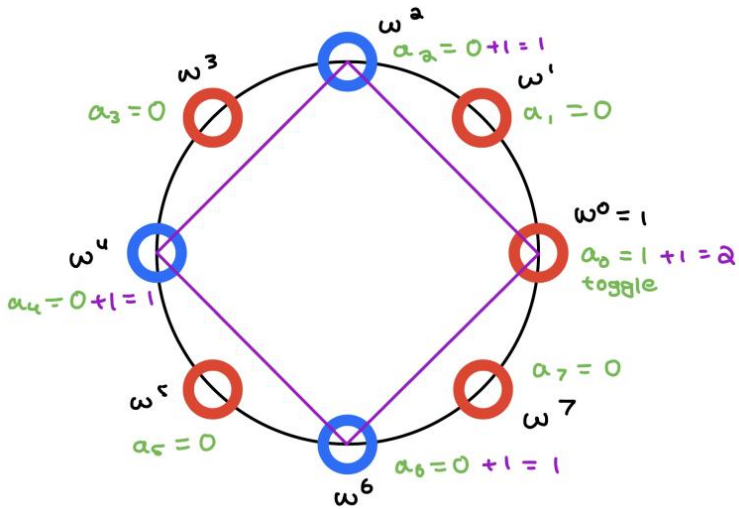
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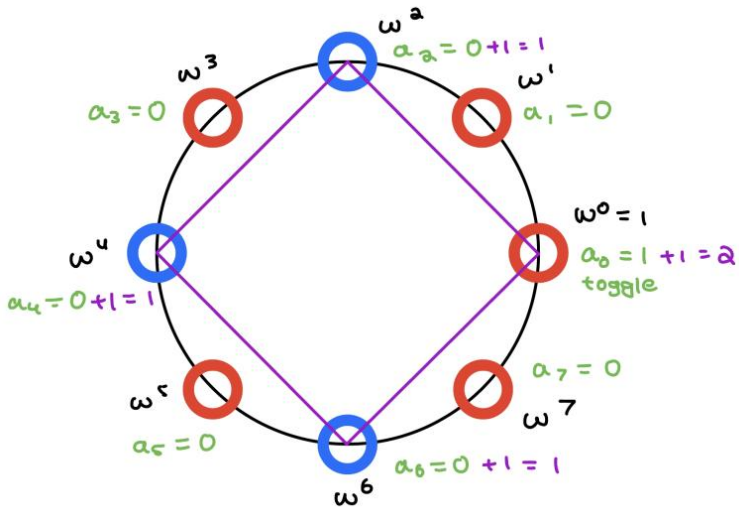
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What's useful now is that $\omega^x + \omega^{x+d} + \ldots + \omega^{x+n-d} = 0$, as we proved before. So $a_0\omega^0 + a_1\omega^1 + \ldots$ never changes! In particular it starts off as equal to $1 + 0 + \ldots = 1$, so thus it always retains that value.

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when all the $a_i$ are *even integers*. To do this, we'll need **algebraic integers.**

# Algebraic Numbers

Recall that a polynomial is an expression of the form $c_n x^n + c_{n-1} x^{n-1} \dots c_0 x^0$; here the $c_j$s are the *coefficients*. When the $c_j$s are all integers, the roots of the polynomial are *algebraic numbers*.

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## Definition

An algebraic number is any complex number that is the root of a polynomial with integer coefficients.

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- Not every number is algebraic; i.e. $\pi$, $e$ are not.

# Algebraic Integers: Motivation

Every integer is the root of an equation $x + c_0$, where $c_0$ is an integer. Clearly a rational number that *isn't an integer* cannot. But can it be the root of some polynomial with integer coefficients, which satisfies that its first coefficient is 1?

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**Theorem (Corollary of RRT)**

*If a rational number $r$ is the root of some polynomial $x^n + c_{n-1}x^{n-1} + \ldots + c_0$, where the $c_j$s are all integers, then $r$ is an integer.*

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### Proof.

Write $r = p/q$ in its lowest terms. If $r$ is a root of $x^n + c_{n-1}x^{n-1} + \ldots + c_0$, then:

$$(p/q)^n + c_{n-1}(p/q)^{n-1} + \ldots + c_0 = 0$$
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But $q$ divides $c^{n-1}p^{n-1}q + \ldots + c_0 q^n$ and yet $q$ does not divide $p^n$, so the LHS is not divisible by $q$. □ □

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An *Algebraic Integer* is a number that is the root of some polynomial $x^n + c_{n-1}x^{n-1} + \ldots + c_0$.

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- $i = \sqrt{-1}$ is an algebraic integer, and in fact so is any $a + bi$, where $a, b \in \mathbb{Z}$.

# Algebraic Integers: Examples

- Integers are algebraic integers, but the other rationals are *not*.
- $\sqrt{2}$ is a root of $x^2 - 2$, and hence an algebraic integer. But $\sqrt{2/3}$ is *not* an algebraic integer. In particular if $n$ is rational, then $\sqrt{n}$ is an algebraic integer iff $n$ is an integer.
- $i = \sqrt{-1}$ is an algebraic integer, and in fact so is any $a + bi$, where $a, b \in \mathbb{Z}$.
- All roots of unity are algebraic integers, as they're all roots of polynomials of the form $x^m - 1$.

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Say $a$ is the root of $x^n + c_{n-1}x^{n-1} + \ldots + c_0$. Then for any integer $m$, $ma$ is the root of $(x/m)^n + c_{n-1}(x/m)^{n-1} + \ldots + c_0$; hence also $x^n + c^{n-1}mx^{n-1} + \ldots + c_0 m^n$. Hence $ma$ is also an algebraic integer. □

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**Theorem**

*If $x$ and $y$ are both algebraic integers, so is $x + y$.*

There's no "easy" proof for this, so we'll just take this as a fact today.

# Solving the Reduced Problem

We want to see if there's a solution to:

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But since all the $b_j$s are integers and all the $\omega^j$s are $n$th roots of unity and hence also algebraic integers, the entire LHS is an algebraic integer. But $\frac{1}{2}$ is not an algebraic integer, and so this is impossible!

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But since all the $b_j$s are integers and all the $\omega^j$s are $n$th roots of unity and hence also algebraic integers, the entire LHS is an algebraic integer. But $\frac{1}{2}$ is not an algebraic integer, and so this is impossible! Hence the answer to the original question is that **it's not possible for all $n$.**

# Algebraic Integers: Beyond

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Algebraic integers and rationals are a good way of generalizing this idea, in a way that preserve a lot of the old structure. Like with integers, we still have analogues of "dividing" and "prime factoring" (cf ideals and ideal factorization), but now factoring and equations in general have become much more flexible. In particular, among other results, algebraic integers gave some of the first few proofs of Fermat's Last Theorem for small $p$.

# Further Reading

- Before you learn more about algebraic integers, I would recommend learning some *linear algebra*. Sheldon Axler's "Linear Algebra Done Right" is a great text for this.

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- The material I taught you today can largely be found in Chapter 15 of Michael Artin's "Algebra". It's a long road to get there; but said book is fantastic for learning about groups, rings, and fields; the last of which is what a good amount of today is about.

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- If you want to delve further into algebraic number theory, a good text would be Daniel A. Marcus' "Number Fields".

# Questions?